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Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester -5

2520030142
N. Siri Chandana

Cloud Infrastructure and Services

25CS2206E

Practical

2026

Experiment -1

AIM: 2 a) Enabling Multi-Factor Authentication to Secure Your Access

Description:

Multi-Factor Authentication (MFA) is a security feature that provides an additional layer of protection to an AWS account. It requires users to verify their identity using a password and a second authentication method, such as an authenticator app or security key.

Steps:

1 Sign in to the AWS Management Console.

2. Open the account menu and select Security Credentials.

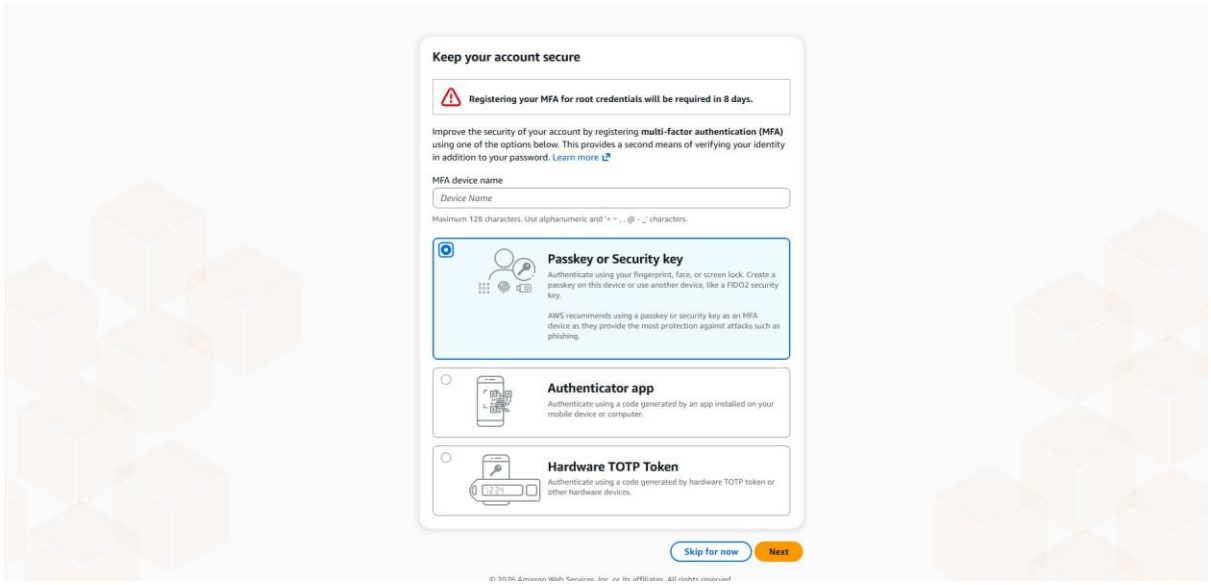
3. Navigate to the Multi-Factor Authentication (MFA) section.

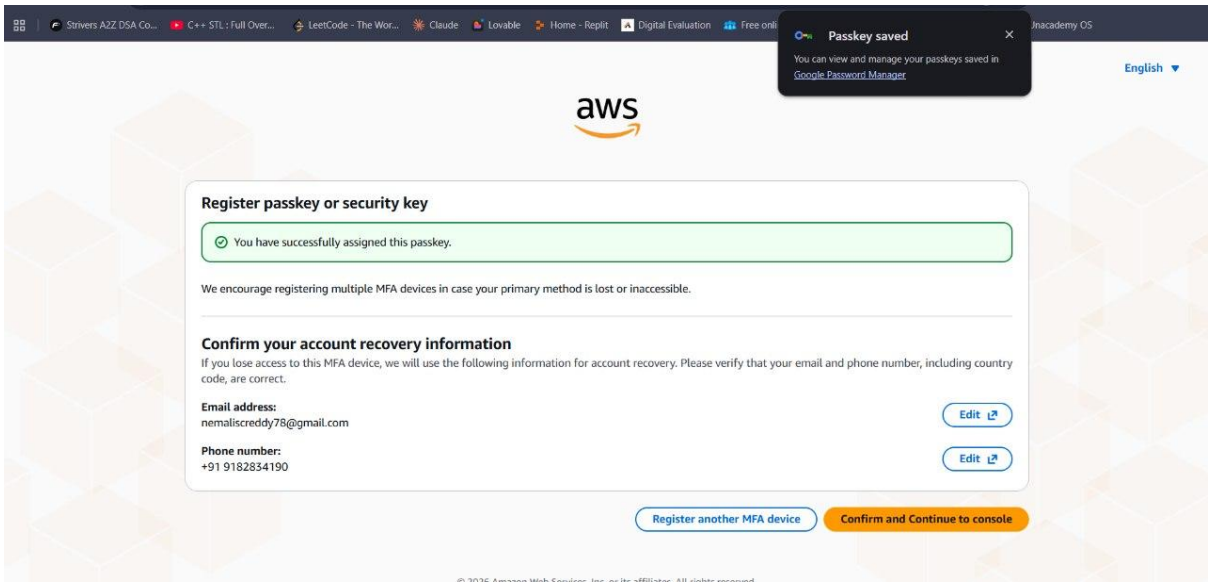
4. Click Assign MFA device and enter a device name.

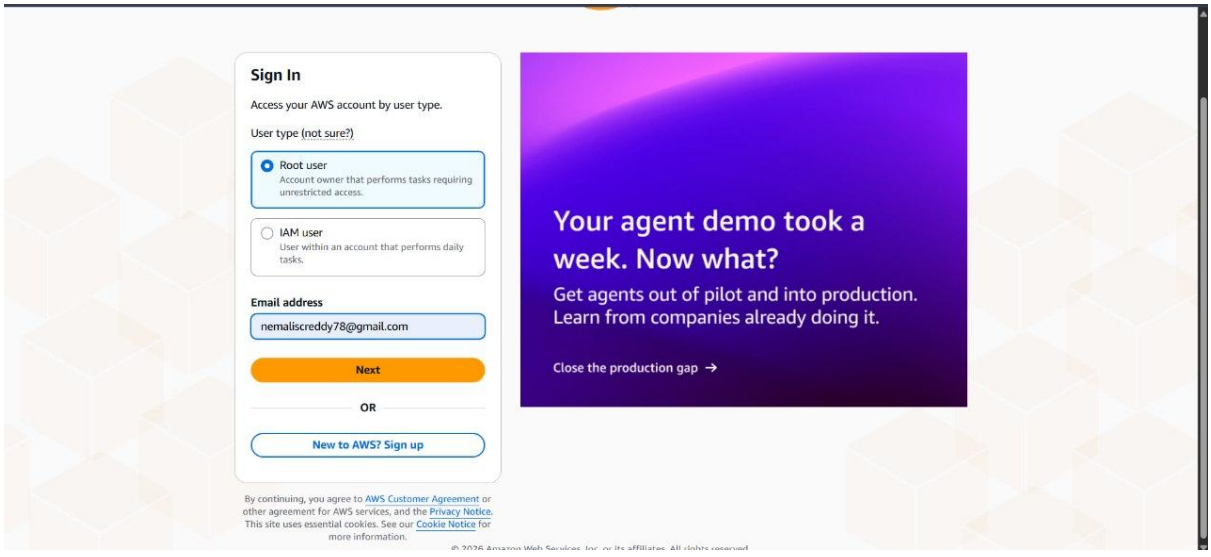
5. Select an authentication method, such as an authenticator app, and scan the QR code.

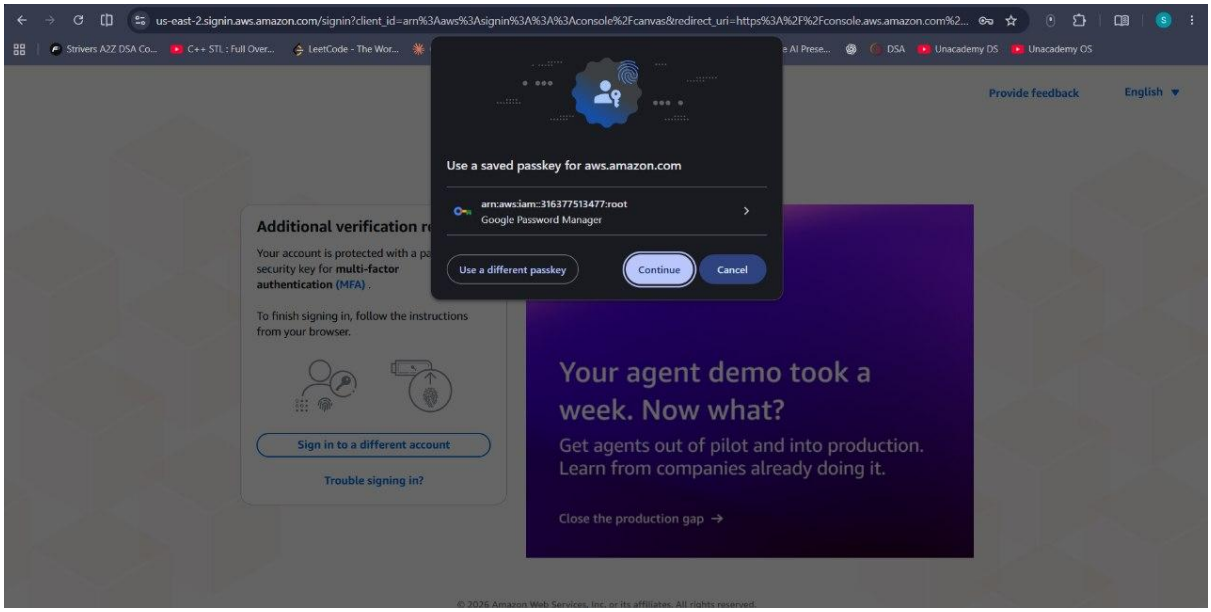
6. Enter the two consecutive MFA codes generated by the app and click Add MFA.

Output:


Keep your account secure
Registering your MFA for root credentials will be required in 8 days.
Improve the security of your account by registering multi-factor authentication (MFA) using one of the options below. This provides a second means of verifying your identity in addition to your password. Learn more
MFA device name
Device Name
Maximum 128 characters. Use alphanumeric and '+ = , . @ - _' characters.
Passkey or Security key
Authenticate using your fingerprint, face, or screen lock. Create a passkey on this device or use another device, like a FIDO2 security key.
AWS recommends using a passkey or security key as an MFA device as they provide the most protection against attacks such as phishing.
Authenticator app
Authenticate using a code generated by an app installed on your mobile device or computer.
Hardware TOTP Token
Authenticate using a code generated by hardware TOTP token or other hardware devices.
Skip for now
Next


Passkey saved
You can view and manage your passkeys saved in Google Password Manager
English
aws
Register passkey or security key
You have successfully assigned this passkey.
We encourage registering multiple MFA devices in case your primary method is lost or inaccessible.
Confirm your account recovery information
If you lose access to this MFA device, we will use the following information for account recovery. Please verify that your email and phone number, including country code, are correct.
Email address:
nemaliscreddy78@gmail.com
Edit
Phone number:
+91 9182834190
Edit
Register another MFA device
Confirm and Continue to console


Sign In
Access your AWS account by user type.
User type (not sure?)
Root user
Account owner that performs tasks requiring unrestricted access.
IAM user
User within an account that performs daily tasks.
Email address
nemaliscreddy78@gmail.com
Next
OR
New to AWS? Sign up
By continuing, you agree to AWS Customer Agreement or other agreement for AWS services, and the Privacy Notice. This site uses essential cookies. See our Cookie Notice for more information.
Your agent demo took a week. Now what?
Get agents out of pilot and into production. Learn from companies already doing it.
Close the production gap →


us-east-2.signin.aws.amazon.com/signin?client_id=arn%3Aaws%3Asignin%3A%3A%3Aconsole%2Fcanvas&redirect_uri=https%3A%2F%2Fconsole.aws.amazon.com%2...
Strivers A2Z DSA Co...
C++ STL : Full Over...
LeetCode - The Wor...
DSA
Unacademy DS
Unacademy OS
Provide feedback
English
Use a saved passkey for aws.amazon.com
arn:aws:iam::316377513477:root
Google Password Manager
Use a different passkey
Continue
Cancel
Additional verification r
Your account is protected with a pa
security key for multi-factor
authentication (MFA) .
To finish signing in, follow the instructions
from your browser.
Sign in to a different account
Trouble signing in?
Your agent demo took a
week. Now what?
Get agents out of pilot and into production.
Learn from companies already doing it.
Close the production gap →
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AIM: 2 b) Creating First Linux Instance

Description:

Amazon EC2 (Elastic Compute Cloud) is an AWS service that provides virtual servers called instances. A Linux instance allows users to run Linux-based applications and perform various computing tasks in the cloud**.**

Steps:

1.Sign in to the AWS Management Console and open the EC2 service.

2. Click on Launch Instance to create a new virtual machine.

3. Enter an instance name and select an Amazon Linux AMI.

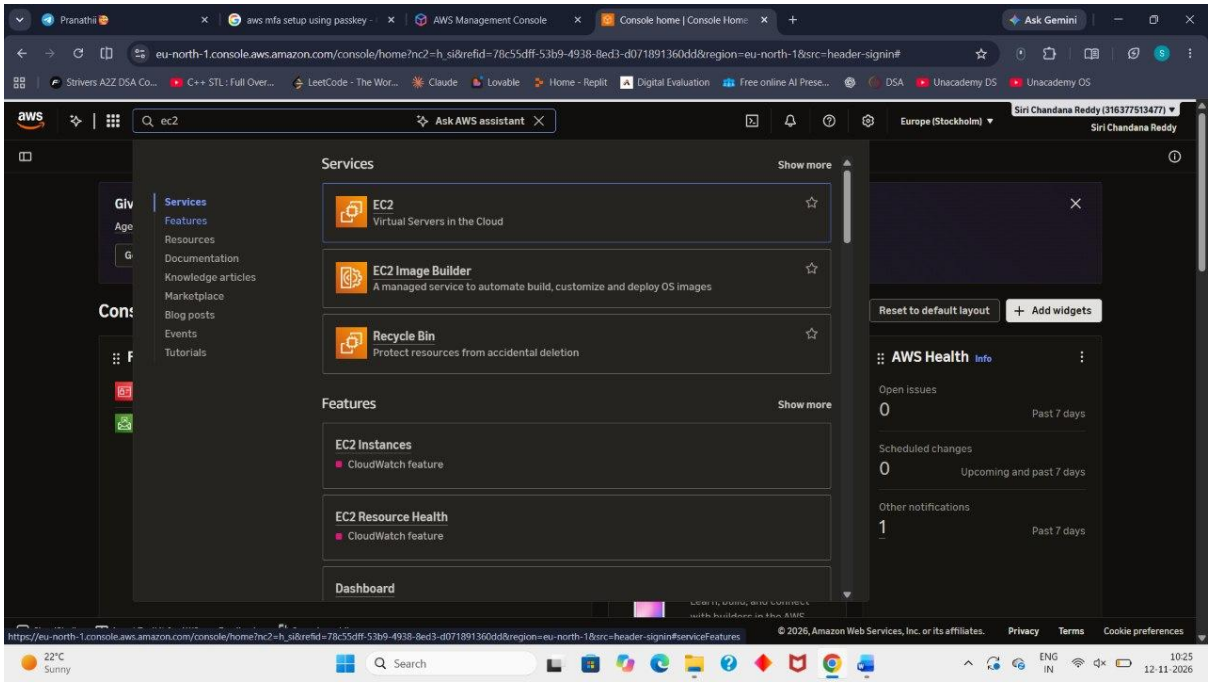
4. Choose an instance type, such as t2.micro or an eligible free-tier instance type.

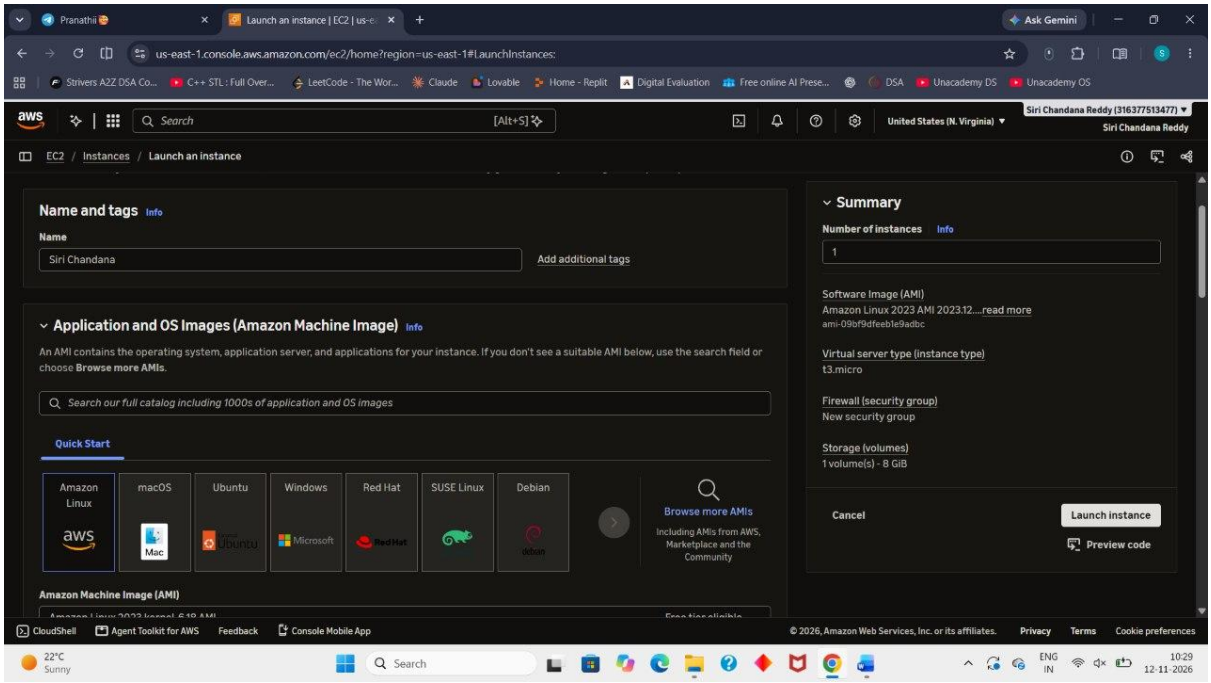
5. Create or select a key pair to securely connect to the instance.

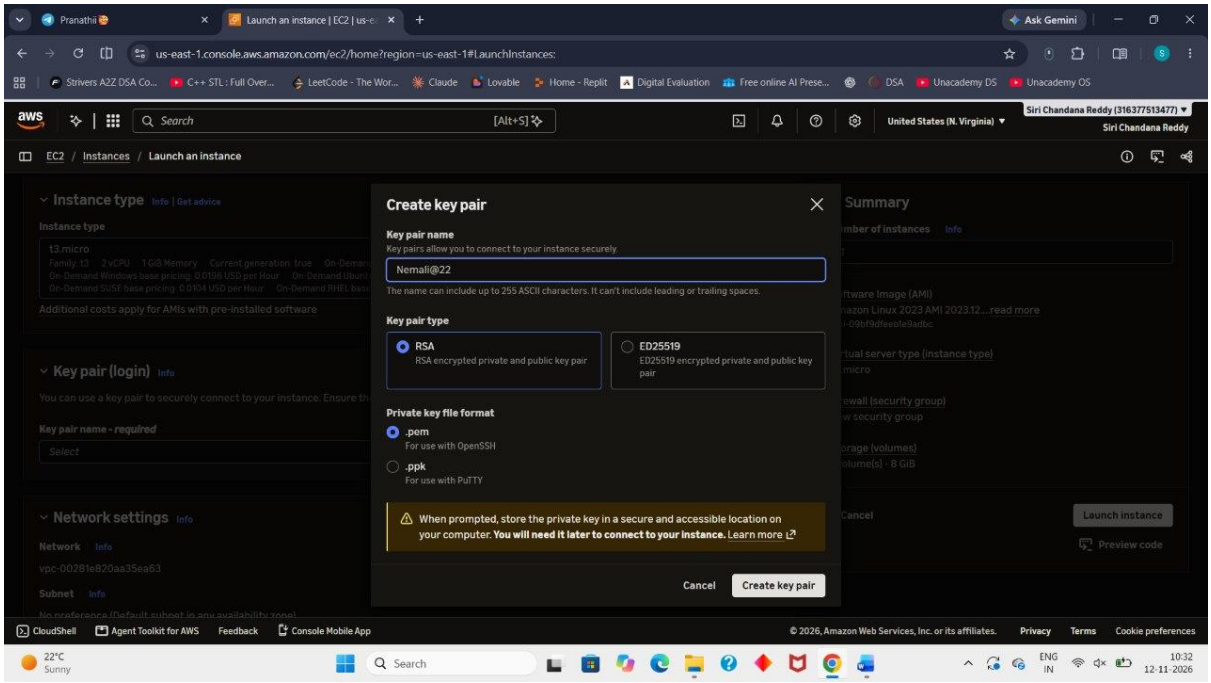
6. Configure network settings and allow SSH traffic (port 22).

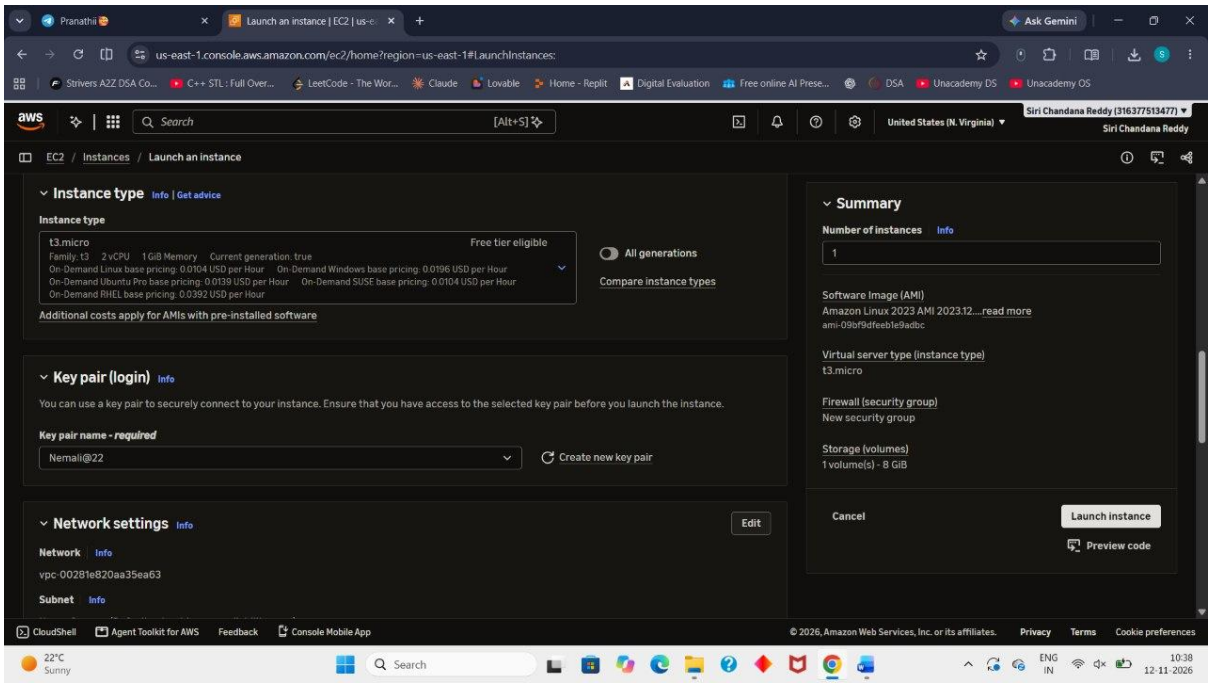
7. Click Launch Instance and wait until the instance state changes to Running.

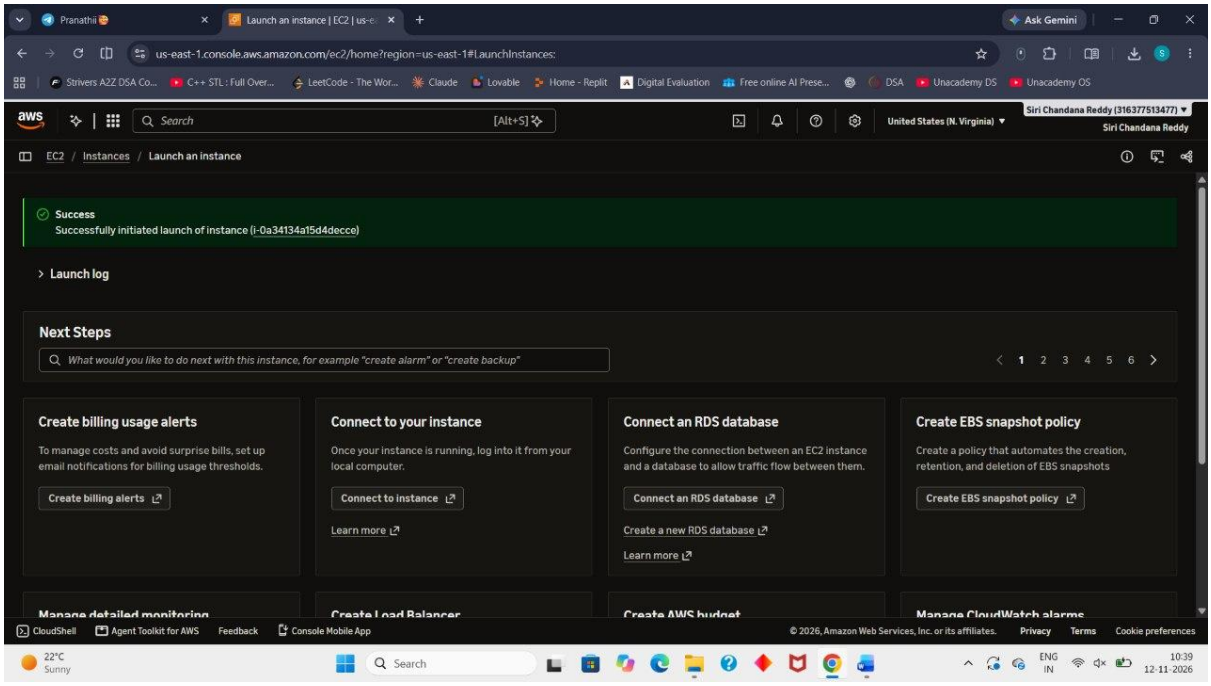
Output:

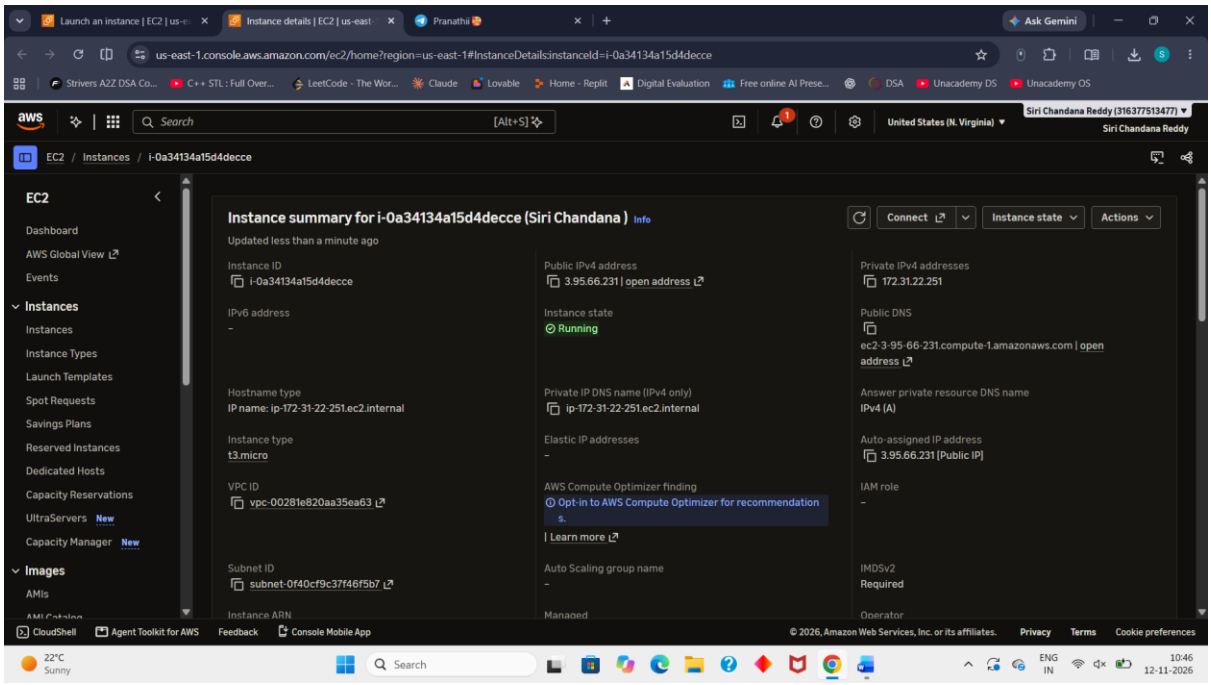

eu-north-1.console.aws.amazon.com/console/home?nc2=h_si&refid=78c55dff-53b9-4938-8ed3-d071891360dd&region=eu-north-1&src=header-signin#
ec2
Ask AWS assistant
Europe (Stockholm)
Siri Chandana Reddy
Services
Show more
EC2
Virtual Servers in the Cloud
EC2 Image Builder
A managed service to automate build, customize and deploy OS images
Recycle Bin
Protect resources from accidental deletion
Features
EC2 Instances
CloudWatch feature
EC2 Resource Health
CloudWatch feature
Dashboard
Reset to default layout
Add widgets
AWS Health Info
Open issues
0
Past 7 days
Scheduled changes
0
Upcoming and past 7 days
Other notifications
1
Past 7 days
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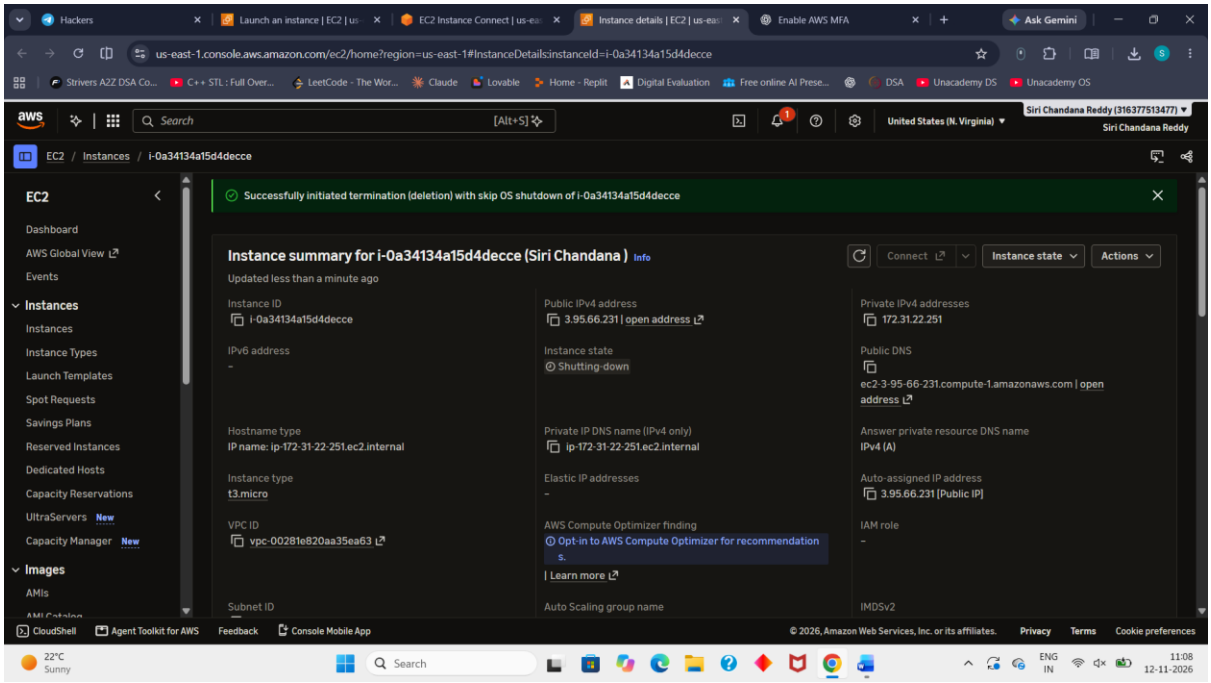

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Alt+S]
United States (N. Virginia)
Siri Chandana Reddy
EC2 / Instances / Launch an instance
Name and tags Info
Name
Siri Chandana
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-09bf9dfeeb1e9adbc
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Launch an instance | EC2 | us-e
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Alt+S]
United States (N. Virginia)
Siri Chandana Reddy (316377513477)
Siri Chandana Reddy
EC2 / Instances / Launch an instance
Instance type
Key pair (login)
Network settings
Summary
Create key pair
Key pair name
Key pairs allow you to connect to your instance securely.
Nemali@22
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Launch an instance | EC2 | us-e
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Alt+S]
United States (N. Virginia)
Siri Chandana Reddy (316377513477)
Siri Chandana Reddy
EC2 / Instances / Launch an instance
Instance type
Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0104 USD per Hour
On-Demand Windows base pricing: 0.0196 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour
On-Demand SUSE base pricing: 0.0104 USD per Hour
On-Demand RHEL base pricing: 0.0392 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Nemali@22
Create new key pair
Network settings Info
Edit
Network Info
vpc-00281e820aa35ea63
Subnet Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12....read more
ami-09bf9dfeeb1e9adbc
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Launch an instance | EC2 | us-e
Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Alt+S]
United States (N. Virginia)
Siri Chandana Reddy (316377513477)
Siri Chandana Reddy
EC2 / Instances / Launch an instance
Success
Successfully initiated launch of instance (i-0a34134a15d4decce)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Create Load Balancer
Create AWS budget
Manage CloudWatch alarms
CloudShell
Agent Toolkit for AWS
Feedback
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Sunny
Search
ENG
IN
10:39
12-11-2026


Launch an instance | EC2 | us-e
Instance details | EC2 | us-east-
Pranathii
Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0a34134a15d4decce
Search
[Alt+S]
United States (N. Virginia)
Siri Chandana Reddy (316377513477)
Siri Chandana Reddy
EC2 / Instances / i-0a34134a15d4decce
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
Instance summary for i-0a34134a15d4decce (Siri Chandana) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0a34134a15d4decce
Public IPv4 address
3.95.66.231 | open address
Private IPv4 addresses
172.31.22.251
IPv6 address
-
Instance state
Running
Public DNS
ec2-3-95-66-231.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-22-251.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-22-251.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
-
Auto-assigned IP address
3.95.66.231 [Public IP]
VPC ID
vpc-00281e820aa35ea63
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
IAM role
-
Subnet ID
subnet-0f40cf9c37f46f5b7
Auto Scaling group name
-
IMDSv2
Required
Instance ARN
Managed
Operator
CloudShell
Agent Toolkit for AWS
Feedback
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12-11-2026


Hackers
Launch an instance | EC2 | us-
EC2 Instance Connect | us-ea
Instance details | EC2 | us-ea
Enable AWS MFA
Ask Gemini
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=i-0a34134a15d4decce
Strivers A2Z DSA Co...
C++ STL : Full Over...
LeetCode - The Wor...
Claude
Lovable
Home - Replit
Digital Evaluation
Free online AI Prese...
DSA
Unacademy DS
Unacademy OS
aws
Search
[Alt+S]
United States (N. Virginia)
Siri Chandana Reddy (316377513477)
Siri Chandana Reddy
EC2 / Instances / i-0a34134a15d4decce
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
Successfully initiated termination (deletion) with skip OS shutdown of i-0a34134a15d4decce
Instance summary for i-0a34134a15d4decce (Siri Chandana) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0a34134a15d4decce
Public IPv4 address
3.95.66.231 | open address
Private IPv4 addresses
172.31.22.251
IPv6 address
-
Instance state
Shutting-down
Public DNS
ec2-3-95-66-231.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-22-251.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-22-251.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
-
Auto-assigned IP address
3.95.66.231 [Public IP]
VPC ID
vpc-00281e820aa35ea63
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations.
Learn more
IAM role
-
Subnet ID
Auto Scaling group name
IMDSv2
CloudShell
Agent Toolkit for AWS
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12-11-2026

AIM: 2 c) S3 Bucket Creation and Upload Objects
I. Create an S3 Bucket for file storage.
II. Upload 5 objects with different file extensions.

Description:

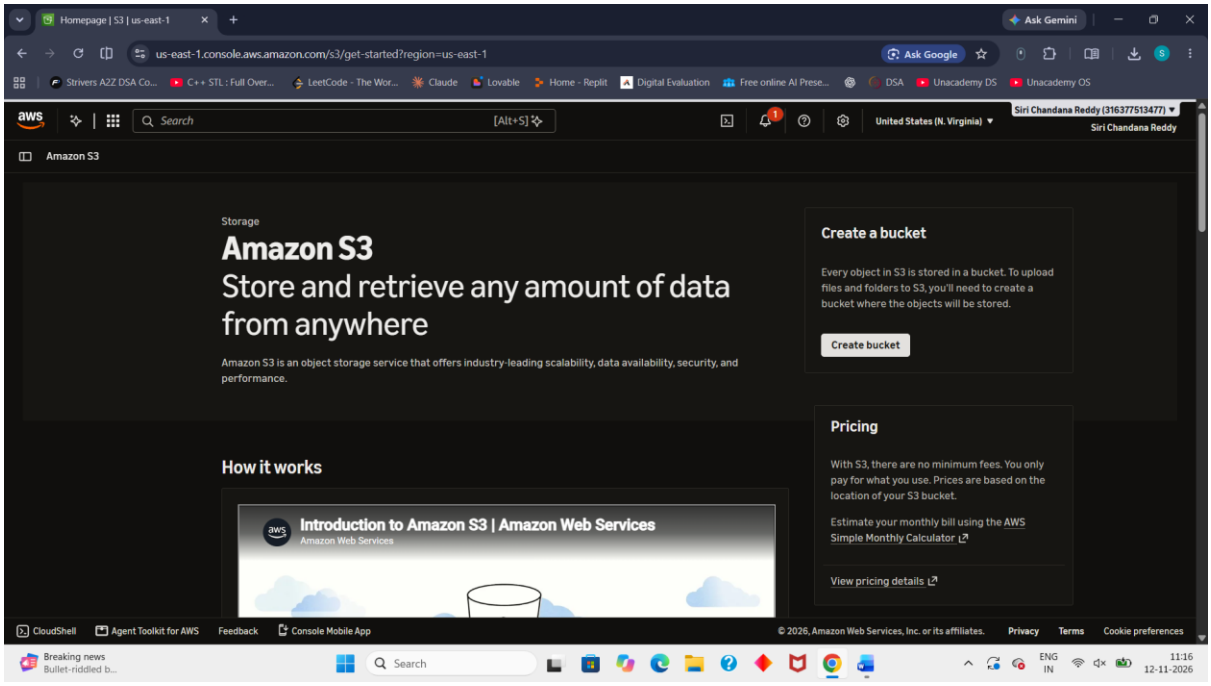
Amazon Simple Storage Service (Amazon S3) is an object storage service provided by AWS. It allows users to store and manage files in containers called buckets. A bucket is created with a globally unique name and an AWS Region, and it can store documents, images, videos and other types of files.

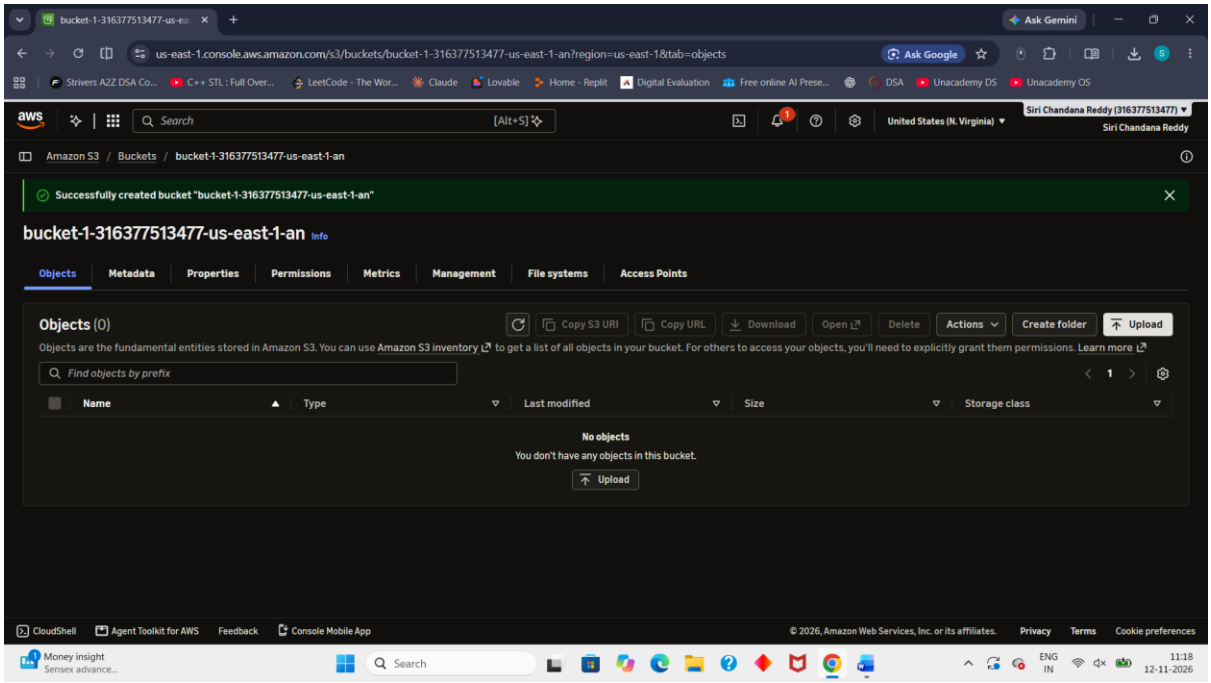
After creating the bucket, users can upload objects in different file formats. In this experiment, five files with different extensions, such as PDF, JPG, PNG, TXT and DOCX, are uploaded. The uploaded objects can be viewed and managed from the S3 bucket.

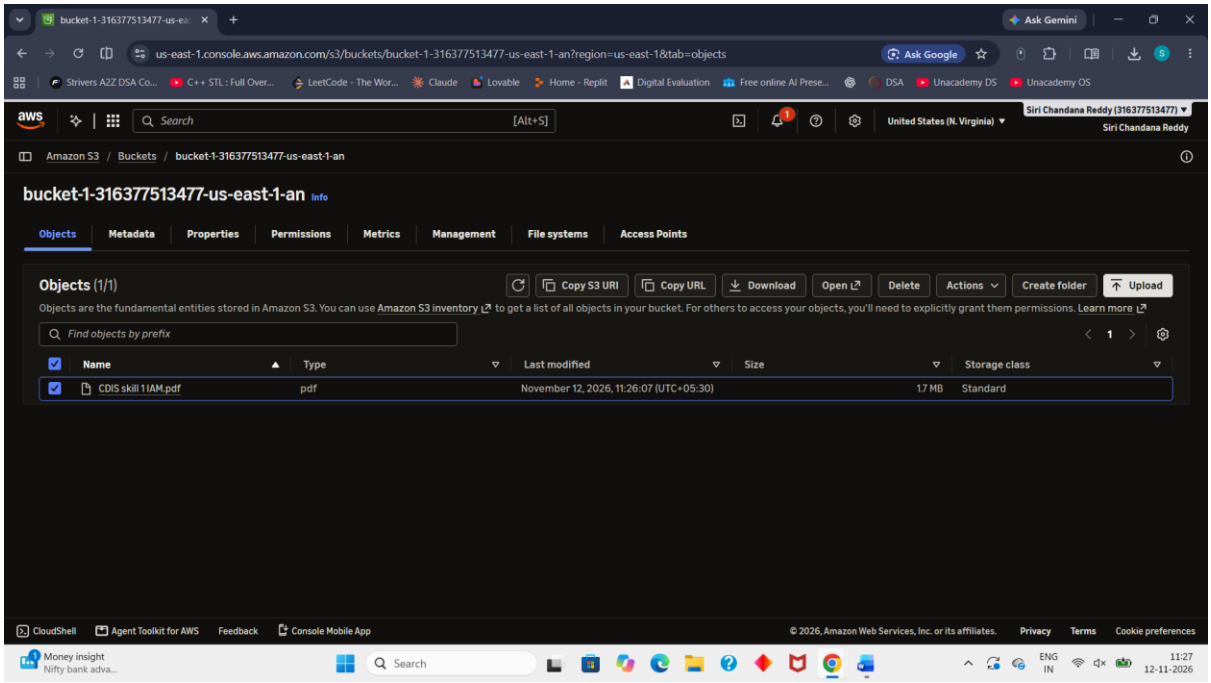
Steps:

1. Sign in to the AWS Management Console and open the Amazon S3 service.
2. Click Create Bucket and enter a unique bucket name.
3. Select the AWS Region and keep the default settings as required.
4. Click Create Bucket to create the storage bucket.
5. Open the created bucket and click Upload.
6. Click Add Files and select five files with different extensions, such as .pdf, .jpg, .txt, .docx, and .png.
7. Click Upload and wait until all five objects are successfully uploaded.

Output:


Homepage | S3 | us-east-1
us-east-1.console.aws.amazon.com/s3/get-started?region=us-east-1
Amazon S3
Storage
Amazon S3
Store and retrieve any amount of data from anywhere
Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.
Create a bucket
Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.
Create bucket
Pricing
With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.
Estimate your monthly bill using the AWS Simple Monthly Calculator
View pricing details
How it works
Introduction to Amazon S3 | Amazon Web Services
Amazon Web Services
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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us-east-1.console.aws.amazon.com/s3/buckets/bucket-1-316377513477-us-east-1-an?region=us-east-1&tab=objects
Amazon S3 / Buckets / bucket-1-316377513477-us-east-1-an
Successfully created bucket "bucket-1-316377513477-us-east-1-an"
bucket-1-316377513477-us-east-1-an Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (0)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
No objects
You don't have any objects in this bucket.
Upload
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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us-east-1.console.aws.amazon.com/s3/buckets/bucket-1-316377513477-us-east-1-an?region=us-east-1&tab=objects
Search
[Alt+S]
United States (N. Virginia)
Siri Chandana Reddy (316377513477)
Siri Chandana Reddy
Amazon S3 / Buckets / bucket-1-316377513477-us-east-1-an
bucket-1-316377513477-us-east-1-an Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (1/1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
CDIS skill 1 IAM.pdf
pdf
November 12, 2026, 11:26:07 (UTC+05:30)
1.7 MB
Standard
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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